TAMS Tower Suitability — Factor Merge Reference

What is merged during Analyze · How 12 scoring factors combine · TAMS open-data screening only

1. Final merge — all 12 factors → one score

When you click **Analyze**, each factor is scored 0–10, then merged into a single site score:

Formula	Detail
Site score (0–10)	Σ (factor_score × weight) ÷ Σ weights
Verdict	≥ 7.0 Preferred · 4.5–7.0 Conditional · < 4.5 Unsuitable

All 12 factors in the merge

#	Factor	Weight	Merge role
1	Terrain slope	**22%**	Shares DEM fetch with #8
2	Road access	**14%**	Standalone (OSRM)
3	Water / flood buffer	**14%**	Feeds #12 corridor check
4	Settlement clearance	**10%**	Feeds #12 corridor check
5	Power connectivity	**10%**	Shares TAMS+OSM with #6, #7
6	Voltage suitability	**8%**	Shares TAMS+OSM with #5, #7
7	Connection distance	**8%**	Shares TAMS+OSM with #5, #6
8	Elevation	**8%**	Shares DEM fetch with #1
9	Land cover	**8%**	Standalone (OSM + Nominatim)
10	Soil / SBC	**8%**	TAMS geotech OR SoilGrids → one factor
11	Wind	**6%**	Standalone (Open-Meteo)
12	Corridor feasibility	**6%**	Merges #1 + #3 + #4 signals

2. Data merges before scoring (collectSiteSignals)

During Analyze, live APIs are fetched in parallel, merged into shared signal buckets, then passed to `scoreSiteSignals()`.

A. DEM → 2 factors (#1 Slope + #8 Elevation)

Source	What is merged	Feeds factors
Open-Meteo Elevation API	5-point grid: centre pad + N/S/E/W (~133 m offset). Centre → elevation. Max angle to offsets → slope.	#1 Terrain slope · #8 Elevation

B. TAMS GIS + OSM Overpass → 3 factors (#5, #6, #7)

Source	Assets merged	Feeds factors
TAMS GIS	Towers, substations, transmission lines	#5 Power connectivity #6 Voltage suitability #7 Connection distance
OSM Overpass	Towers, portals, poles (≤15 km), lines, cables, substations, plants	

All assets are deduplicated into one **nearbyPower** list. From that single merge:

- **#5** — nearest asset distance + bonuses (pole ≤350 m, tower ≤2 km, substation, interconnect ease)
- **#6** — voltage tags from merged assets vs planning tiers (11–765 kV)

- **#7** — practical spur = nearest distance × 1.2

C. TAMS Geotech OR SoilGrids → 1 factor (#10)

Priority	Source	Rule
1 (preferred)	TAMS field geotech within ~5 km	Uses adopted SBC, CBR, design depth from /geotech
2 (fallback)	ISRIC SoilGrids 2.0	Texture class, indicative SBC/CBR, ~40–48% screening confidence

Only one path is used — field geotech wins when present; otherwise open GIS soil screening.

D. Water + Settlement + Slope → 1 factor (#12)

Factor #12 does **not** fetch new data. It reuses signals already computed for other factors:

Reused signal	From factor	Penalty if triggered
Water < 150 m	#3 Water / flood buffer	−2
Settlement < 100 m	#4 Settlement clearance	−2
Slope > 12°	#1 Terrain slope	−1.5
Slope > 18°	#1 Terrain slope	−1

Starts at score 8; deductions applied; capped 0–10.

3. Standalone factors (no merge with others)

#	Factor	Source only
2	Road access	OSRM nearest drivable road
3	Water / flood buffer	OSM water features (+ Photon fallback if Overpass fails)
4	Settlement clearance	OSM buildings/places (+ Photon fallback)
9	Land cover	OSM landuse/natural + Nominatim fallback
11	Wind exposure	Open-Meteo 90-day mean daily max wind @ 10 m

4. Confidence % — partial merge (not all 12)

Confidence is **not** calculated from all 12 factors. It counts **7 resolved signals**:

Included in confidence	Maps to factor
Elevation resolved	#8
Slope resolved	#1
Road distance	#2
Water distance	#3
Settlement distance	#4
Grid (tower or substation)	#5 (partial)
Wind resolved	#11

Not in confidence directly: #6 Voltage · #7 Connection distance · #9 Land cover · #10 Soil · #12 Corridor feasibility

Formula: base 55% + up to 25% (resolved ÷ 7 × 25) − 5% per Photon/fallback used (water, settlement, grid unavailable).

5. NOT merged into the 12-factor score

Module	Runs at analyze?	In /10 score?
Geotechnical Intelligence (GEO-1 Word/HTML report)	Yes	No — additive block outside scoreSiteSignals
Corridor placement panel (Suggest / Shift / Skip / Review per pad)	Yes (KML/line)	No — separate CEA span screening
Power network verdict (Yes / No / Unknown)	Yes	No — informational only
OSRM road route overlay (orange line on map)	On demand	No — display only

6. Analyze flow diagram

```
Analyze (collectSiteSignals)
  │
  ├─ Parallel fetch
  │    ├─ Open-Meteo DEM (5 pts)     ──→ MERGE → #1 Slope + #8 Elevation
  │    ├─ OSRM nearest road          ──→ #2 Road (standalone)
  │    ├─ OSM water (+ Photon fb)    ──→ #3 Water (+ feeds #12)
  │    ├─ OSM settlement (+ Photon)  ──→ #4 Settlement (+ feeds #12)
  │    ├─ OSM landuse + Nominatim    ──→ #9 Land cover (standalone)
  │    ├─ Open-Meteo wind            ──→ #11 Wind (standalone)
  │    ├─ TAMS + OSM power assets    ──→ MERGE → #5 + #6 + #7
  │    ├─ TAMS geotech OR SoilGrids   ──→ MERGE → #10 (field wins)
```

```
│    └─ Nominatim place label      ──► header only
│
├─ scoreSiteSignals()
│    ├─ Score each factor 0–10
│    ├─ #12 merges water + settlement + slope (reuse)
│    └─ MERGE ALL 12 → finalScore → verdict
│
└─ Separate (not in score)
     ├─ Geotech Word report
     └─ Corridor placement advice (T1…Tn)
```

7. Quick reference — merge groups

Merge group	Factors affected	Shared data
DEM group	#1, #8	Open-Meteo 5-point elevation grid
Power group	#5, #6, #7	TAMS + OSM deduped nearbyPower
Soil group	#10	TAMS geotech OR SoilGrids (one path)
Corridor group	#12 (uses #1, #3, #4)	Water km + building km + slope °
Final score	All #1–#12	Weighted average → verdict

> *This document describes how TAMS Tower Suitability merges open-data signals during Analyze. It is screening logic only — not utility approval, ROW certification, or foundation design.*